

SCHOOL OF ADVANCED SCIENCES

Winter Semester 2023-2024

Continuous Assessment Test –II

Programme Name & Branch : B.Tech (All)

Slot: B1+TB1-Common

Course Name & code: Probability and Statistics & BMAT202L

Class Number (s): VL2023240501675, VL2023240504939, VL2023240501741,
VL2023240501679, VL2023240501681, VL2023240501677, VL2023240501674, VL2023240501739,
VL2023240501683, VL2023240501685

Exam Duration: 90 Min.

Maximum Marks: 50

General instruction(s): Answer ALL Questions

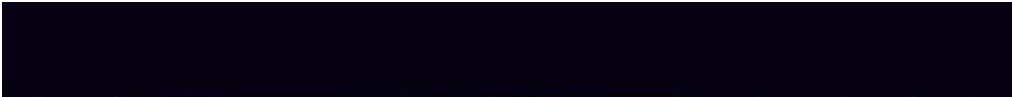
(Use of statistical table is allowed)

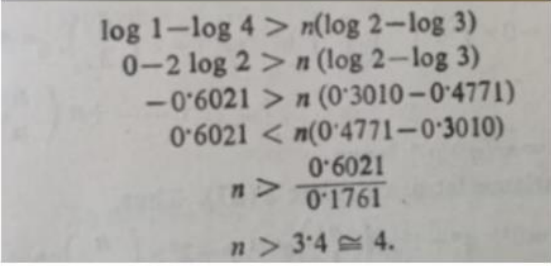
Q.No.	Question	Max Marks	CO	BL																																	
1.	The following table gives the test scores made by ten sales man on an intelligent test and the values of their weekly sales: <table><tr><td>Salesman</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td><td>G</td><td>H</td><td>I</td><td>J</td></tr><tr><td>Test scores</td><td>40</td><td>70</td><td>50</td><td>60</td><td>80</td><td>50</td><td>90</td><td>40</td><td>60</td><td>60</td></tr><tr><td>Sales (in thousands)</td><td>2.5</td><td>6.0</td><td>4.5</td><td>5.0</td><td>4.5</td><td>2.0</td><td>5.5</td><td>3.0</td><td>4.5</td><td>3.0</td></tr></table> Obtain the regression equation of sales on test scores. Also estimate the most probable value of weekly sales for a salesman whose test score is 70.	Salesman	A	B	C	D	E	F	G	H	I	J	Test scores	40	70	50	60	80	50	90	40	60	60	Sales (in thousands)	2.5	6.0	4.5	5.0	4.5	2.0	5.5	3.0	4.5	3.0	10	CO	BL
Salesman	A	B	C	D	E	F	G	H	I	J																											
Test scores	40	70	50	60	80	50	90	40	60	60																											
Sales (in thousands)	2.5	6.0	4.5	5.0	4.5	2.0	5.5	3.0	4.5	3.0																											

Solution.				
Test Scores	Sales	$\frac{x - 50}{10}$	$\frac{y - 4.5}{(0.5)}$	

	$\sigma_x^2 = \frac{34}{10} - \left(\frac{10}{10}\right)^2 = 2.4$ $\sigma_y^2 = \frac{73}{10} - \left(\frac{-9}{10}\right)^2 = 6.49$ Thus, $r = \frac{2.8}{(1.549)(2.547)} = 0.71$ $\sigma_x = 1.549 \times 10 = 15.49$ $\sigma_y = 2.547 \times \frac{5}{10} = 1.27$ $\bar{x} = 50 + 10 \times 1 = 60$ $\bar{y} = 4.5 + \frac{5}{10}(-0.9) = 4.05$ Thus, the regression of sales (y) on test score (x) is $Y_c - \bar{y} = r \frac{\sigma_y}{\sigma_x} (X - \bar{x})$ which reduces to $Y_c - 4.05 = (0.71) \left(\frac{1.27}{15.49} \right) (X - 60)$ That is, $Y_c = 0.558 + 0.582 X$ Hence, an estimate of weekly sales for a test score of 70 is $Y_c = 0.558 + (0.582)(70) = 41.632 \text{ thousand rupees} = \text{Rs. } 41632.$			
2.	The probability of a man scoring a penalty in a Hockey match is $\frac{1}{3}$. How many times he should be given a chance so that the probability of scoring successfully at least once is greater than $\frac{3}{4}$.	10	CO	BL

	Solution. Let p be the probability of scoring a penalty corner and n be the number of chances given. Given $p = \frac{1}{3}$ $\therefore q = 1 - \frac{1}{3} = \frac{2}{3}$ Probability of scoring successfully at least once is $1 - q^n = 1 - \left(\frac{2}{3}\right)^n$ We find n such that $1 - \left(\frac{2}{3}\right)^n > \frac{3}{4}$ or $\frac{1}{4} > \left(\frac{2}{3}\right)^n$ or $\log \frac{1}{4} > n \log \frac{2}{3}$ $\log 1 - \log 4 > n(\log 2 - \log 3)$ $0 - 2 \log 2 > n(\log 2 - \log 3)$ $-0.6021 > n(0.3010 - 0.4771)$ $0.6021 < n(0.4771 - 0.3010)$ $n > \frac{0.6021}{0.1761}$ $n > 3.4 \approx 4.$			
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	 $\begin{aligned}\log 1 - \log 4 &> n(\log 2 - \log 3) \\ 0 - 2 \log 2 &> n(\log 2 - \log 3) \\ -0.6021 &> n(0.3010 - 0.4771) \\ 0.6021 &< n(0.4771 - 0.3010) \\ n &> \frac{0.6021}{0.1761} \\ n &> 3.4 \cong 4.\end{aligned}$			
3.	If the number of kilometers that a car can run before its battery wears out is exponentially distributed with an average value of 10,000 km and if the owner desires to take a 5000 km trip, what is the probability that he will be able to complete the trip without having to replace the car battery? Assume that the car has been used for some time. The given problem can be solved using the exponential distribution formula. Let X be the number of miles the car can run before its battery wears out, and λ be the rate parameter of the exponential distribution. The average value of X is given as 10,000 miles, so we have: $E(X) = 10,000$ We know that the mean of an exponential distribution is equal to the inverse of the rate parameter. Therefore, we can write: $E(X) = 1/\lambda$ Substituting the value of $E(X)$, we get: $10,000 = 1/\lambda$	10	CO	BL



	parameter $1/10$ (in thousands of miles), $P(\text{remaining lifetime} > 5) = 1 - F(5) = e^{-5/10} = e^{-1/2} \approx .604$ If the lifetime distribution is not exponential, $P(\text{lifetime} > t + 5 \text{lifetime} > t) = \frac{P(\text{lifetime} > t + 5, \text{lifetime} > t)}{P(\text{lifetime} > t)}$ $= \frac{1 - F(t + 5)}{1 - F(t)}$															
4.	A cool drinks manufacturing company claims that its brand A cool drinks outsells its brand B by 8%. It is found that 42 out of a sample of 200 people prefer brand A and 18 out of another sample of 100 people prefer brand B. Test at 10% level of significance, whether the 8% difference is a valid claim. Ans: The test statistic of Z is $z = 1.021$ whereas $z_\alpha = 1.645$ Hence, the 8% difference in the sale of two brands of products is a valid claim by the firm.	10	CO	BL												
5.	Two samples drawn from two different populations gave the following results: <table><tr><th>Sample</th><th>Size</th><th>Mean</th><th>SD</th></tr><tr><td>I</td><td>400</td><td>124</td><td>14</td></tr><tr><td>II</td><td>250</td><td>120</td><td>12</td></tr></table>	Sample	Size	Mean	SD	I	400	124	14	II	250	120	12	10	CO	BL
Sample	Size	Mean	SD													
I	400	124	14													
II	250	120	12													



Test the significance of the difference between the means of the samples and also find the 99% confidence limits for the difference of the population means. Answer: Null Hypothesis (H_0): There is no significant difference between the means of the two populations. Alternative Hypothesis (H_1): There is a significant difference between the means of the two			
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Programme Name & Branch : B. Tech. (Common)
Course Code and Course Name : BMAT202L-Probability and Statistics
Faculty Name(s) : Common Slot QP
Class Number(s) : Common Slot QP
Date of Examination : 22/03/2025
Exam Duration : 90 minutes **Maximum Marks: 50**

General instruction(s):

- Answer All Questions.
- Use of statistical table is permitted.
- M - Max mark; CO - Course Outcome; BL - Blooms Taxonomy Level (1 - Remember, 2 - Understand, 3 - Apply, 4 - Analyse, 5 - Evaluate, 6 - Create)
- Course Outcomes: CO2. Understand the basic concepts of random variables and find an appropriate distribution for analyzing data specific to an experiment.
CO3. Apply statistical methods like correlation, regression analysis in analyzing, interpreting experimental data.
CO4. Make appropriate decisions using statistical inference that is the central to experimental research.

Q. No.	Question	Max Marks	CO	BL																						
1.	Find the regression equations for the following data and estimate the values of Y for X=3, 5 and 19: <table><tr><td>X</td><td>2</td><td>6</td><td>4</td><td>12</td><td>10</td><td>8</td><td>0</td><td>18</td><td>14</td><td>16</td></tr><tr><td>Y</td><td>0</td><td>2</td><td>1</td><td>5</td><td>4</td><td>3</td><td>-1</td><td>8</td><td>6</td><td>7</td></tr></table>	X	2	6	4	12	10	8	0	18	14	16	Y	0	2	1	5	4	3	-1	8	6	7	10	3	3
X	2	6	4	12	10	8	0	18	14	16																
Y	0	2	1	5	4	3	-1	8	6	7																
2.	If X is a Poisson variate satisfying $P(X = 2) = 9P(X = 4) + 90P(X = 6).$ Find $P(X < 3)$, $P(X \geq 3)$ and $P(X < 3 X > 0)$.	10	2	3																						
3.	A random variable X follows normal distribution with mean 3 and variance 16. Write the probability density function of X. Also find $P(X \leq 3)$, $P(X - 1 > 2)$ and $P(X < 5 X > 2)$.	10	2	3																						
4.	A sample of 900 members has a mean 3.5 cms and SD 2.61 cms. Is the sample from a large population of mean 3.35 cms and SD 2.61 cms? Test at 2% and 5% LOS. Also, find the 95% confidence interval for the population mean.	10	4	2																						
5.	Random samples of 400 men and 600 women were asked whether they would like to have a flyover near their residence. 225 men and 340 women were in favour of proposal. Test the hypothesis that the proportions of men and women in favour of proposal are same against that they are not, at 5% LOS.	10	4	2																						

Q.No	Question	M	CO	BL																		
1.	The grades of a class of 8 students on internal assessment (X) and on final assessment (Y) are as follows <table><tr><td>X</td><td>22</td><td>26</td><td>29</td><td>30</td><td>31</td><td>31</td><td>34</td><td>35</td></tr><tr><td>Y</td><td>20</td><td>20</td><td>21</td><td>29</td><td>27</td><td>24</td><td>27</td><td>31</td></tr></table> (i) Estimate the linear regression lines. (ii) Estimate the final assessment grade of a student who has scored 38 in the internal assessment. (iii) Find the coefficient of correlation between X and Y.	X	22	26	29	30	31	31	34	35	Y	20	20	21	29	27	24	27	31	10	3	3
X	22	26	29	30	31	31	34	35														
Y	20	20	21	29	27	24	27	31														
2.	a) A pharmaceutical lab states that a drug causes negative side effects in 3 of every 100 patients. To confirm this affirmation, another laboratory chooses 5 people at random who have consumed the drug. What is the probability that (i) none of the five patients experience side effects? (ii) atleast two experience side effects? b) In a component manufacturing industry, there is a small probability of $\frac{1}{500}$ for any component to be defective. The components are supplied in packets of 10. For a consignment of 10,000 packets, use Poisson distribution to calculate the approximate number of packets containing (i) one defective, (ii) atleast one defective.	10	2	2																		

Page 1 o



VIT

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

SCHOOL OF ADVANCED SCIENCES
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

REG.NO.:

SLOT: B2+TB2

CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

SLOT: B2+TB2

3.	In a normal distribution 10% of the items are under 35 and 5% are above 90. Find the mean and standard deviation of the distribution.	10	2	3												
4.	A car manufacturer aims to improve the quality of the products by reducing the defects and also increase the customer satisfaction. Therefore, he monitors the efficiency of two assembly lines in the shop floor. In line A there are 18 defects reported out of 200 samples. While the line B shows 25 defects out of 600 cars. At a 5% significance level, is there a difference between two assembly procedures significant?	10	4	3												
5.	There are two groups of students preparing for a competition: Group A and Group B. Group A has studied offline classes, while Group B has studied online classes. After the examination, based on the score of each student, we have the following observation. <table><tr><td></td><td>Group A</td><td>Group B</td></tr><tr><td>Sample Size</td><td>50</td><td>60</td></tr><tr><td>Sample mean</td><td>75</td><td>80</td></tr><tr><td>Sample Standard Deviation</td><td>10</td><td>12</td></tr></table> Assuming a 5% significance level, perform a two-sample z-test to determine if there is a significant difference between the online and offline classes.		Group A	Group B	Sample Size	50	60	Sample mean	75	80	Sample Standard Deviation	10	12	10	4	3
	Group A	Group B														
Sample Size	50	60														
Sample mean	75	80														
Sample Standard Deviation	10	12														

Q.No.	Question	Marks																											
1.	The following data show the number of bedrooms, the number of baths, and the prices at which a random sample of eight family houses sold recently in a certain large housing development: <table><tr><td>No. of bedrooms</td><td>3</td><td>2</td><td>4</td><td>2</td><td>3</td><td>2</td><td>5</td><td>4</td></tr><tr><td>No. of baths</td><td>2</td><td>1</td><td>3</td><td>1</td><td>2</td><td>2</td><td>3</td><td>2</td></tr><tr><td>Price (dollars)</td><td>78800</td><td>74300</td><td>83800</td><td>74200</td><td>79700</td><td>74900</td><td>88400</td><td>82900</td></tr></table> Use the method of least squares to find a linear equation which will enable us to predict the average sales price of a family house in the given housing development in the terms of the number of bedrooms and the number of baths.	No. of bedrooms	3	2	4	2	3	2	5	4	No. of baths	2	1	3	1	2	2	3	2	Price (dollars)	78800	74300	83800	74200	79700	74900	88400	82900	10
No. of bedrooms	3	2	4	2	3	2	5	4																					
No. of baths	2	1	3	1	2	2	3	2																					
Price (dollars)	78800	74300	83800	74200	79700	74900	88400	82900																					
2.	(a) A multiple choice test consist of 8 question with 3 option of each question in which only one is correct. A student answer each question by rolling a balanced die and checking the first answer if he gets 1 or 2, the second answer if he get 3 or 4 and the third answer if he get 5 or 6. To get a distinction, the student must secure atleast 75%	5+5																											

correct answer. If there is no negative marking, what is the probability that the student secures a distinction?

(b) The mean number of bacteria per millilitre of a liquid is known to be 6. Find the probability that in 1 ml of the liquid, there will be: (a) 0, (b) 1, (c) 2, (d) 3, (e) less than 4, (f) 6 bacteria

	secures a distinction? (b) The mean number of bacteria per millilitre of a liquid is known to be 6. Find the probability that in 1 ml of the liquid, there will be: (a) 0, (b) 1, (c) 2, (d) 3, (e) less than 4, (f) 6 bacteria	
3.	In a sample of 1000 cases, the mean of a certain test is 14 and standard deviation is 2.5. Assuming the distribution to be normal, find (i) how many students score between 12 and 15? (ii) how many score above 18? (iii) how many score below 18?	10
4.	500 articles from a factory are examined and found to be 2% defective. 800 similar articles from a second factory are only found to be 1.5% defective. Can it reasonably be concluded that the products of the first factory are inferior to those of the second?	10
5.	The average marks scored by 32 boys is 72 with a SD of 8, while that for 36 girls is 70 with a SD of 6. Test at 1% LOS whether the boys perform better than girls.	10

Q. No	Question	M	CO	DO																																												
1.	The following are data for the gasoline mileage (y) of an automobile based on its weight (x_1) and temperature (x_2) at the time of operation. Estimate y for $x_1 = 1.2$ and $x_2 = 35$. <table><tr><td>Car Number</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>Miles per gallon (y)</td><td>17.9</td><td>16.5</td><td>16.4</td><td>16.8</td><td>18.8</td><td>15.5</td><td>17.5</td><td>16.4</td><td>15.9</td><td>18.3</td></tr><tr><td>Weights in tons (x_1)</td><td>1.35</td><td>1.90</td><td>1.70</td><td>1.80</td><td>1.30</td><td>2.05</td><td>1.60</td><td>1.80</td><td>1.85</td><td>1.40</td></tr><tr><td>Temperature (x_2)</td><td>90</td><td>30</td><td>80</td><td>40</td><td>35</td><td>45</td><td>50</td><td>60</td><td>35</td><td>30</td></tr></table>	Car Number	1	2	3	4	5	6	7	8	9	10	Miles per gallon (y)	17.9	16.5	16.4	16.8	18.8	15.5	17.5	16.4	15.9	18.3	Weights in tons (x_1)	1.35	1.90	1.70	1.80	1.30	2.05	1.60	1.80	1.85	1.40	Temperature (x_2)	90	30	80	40	35	45	50	60	35	30	10	3	2
Car Number	1	2	3	4	5	6	7	8	9	10																																						
Miles per gallon (y)	17.9	16.5	16.4	16.8	18.8	15.5	17.5	16.4	15.9	18.3																																						
Weights in tons (x_1)	1.35	1.90	1.70	1.80	1.30	2.05	1.60	1.80	1.85	1.40																																						
Temperature (x_2)	90	30	80	40	35	45	50	60	35	30																																						
2.	(a) A machine produces identical units. The proportion of defective units produced by the machine is known to be 1/20. We also know the successive units are statistically independent. Obtain the probability that in a sample of 10 units, there are at most 2 defectives. (b) The probability that a person recovers from a serious disease is 0.40. Find the probability that at least one of the 8 persons admitted to a hospital will survive.	10	2	2																																												
3.	A large group of students took a test in Mathematics and the final score have a mean of 80 and a standard deviation of 15. If the scores are normally	10	2	2																																												

Page 1 of 2



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SCHOOL OF ADVANCED SCIENCES
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

REG.NO.:

SLOT: B1+TB1

	distributed, when percent of the students (i) got a score below 70? (ii) passed the test given that the passing mark is 75? (iii) got a score between 90 and 100?			
4.	In a sample of 600 students of a certain college, 400 are found to use dot pens. In another college from a sample of 900 students 450 were found to use dot pens. Test with 1% LOS, whether the two colleges significantly differ with respect to the habit of using dot pens. Null and alternative hypothesis should be stated clearly.	10	4	3
5.	A bakery claims their new muffins have an average diameter of 5 centimeters (cm) with a standard deviation of 0.5 cm. You suspect the muffins might be smaller. To investigate this, you randomly sample 49 muffins and measure their diameters. The average diameter of your sample is 4.8 cm. Determine if there is enough evidence to suggest the bakery's claim (average diameter of 5 cm) is inaccurate at a 5% significance level. What about 1% level of significance?	10	4	3

1.	A researcher is studying the relationship between sleep duration (in hours) and academic performance (GPA) for a group of 20 students. They collect the following data: <table border="1"> <tr> <th>Student</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> </tr> <tr> <td>Sleep duration</td> <td>4</td> <td>7</td> <td>8</td> <td>9</td> <td>6</td> <td>6.5</td> <td>7.5</td> <td>4.5</td> <td>8.5</td> </tr> <tr> <td>GPA</td> <td>8</td> <td>9</td> <td>9.5</td> <td>8.5</td> <td>8</td> <td>7</td> <td>5</td> <td>6</td> <td>6.5</td> </tr> </table> Are the sleep duration and GPA of the students consistent?	Student	1	2	3	4	5	6	7	8	9	Sleep duration	4	7	8	9	6	6.5	7.5	4.5	8.5	GPA	8	9	9.5	8.5	8	7	5	6	6.5			
Student	1	2	3	4	5	6	7	8	9																									
Sleep duration	4	7	8	9	6	6.5	7.5	4.5	8.5																									
GPA	8	9	9.5	8.5	8	7	5	6	6.5																									
2.	(a) A machine produces identical units. The proportion of defective units produced by the machine is known to be $1/20$. We also know that successive units are statistically independent. Obtain the probability that in a sample of 10 units, there are at most 2 defectives. (b) The probability that a person recovers from a serious	10	CO3	BL3																														

	disease is 0.40. Find the probability that at least one of the 8 persons admitted to a hospital will survive.			
3.	The height of female students at a University follows approximately a normal distribution, with mean 60 inches and standard deviation 2. Find the probability that a female student selected at random has height a) less than 58 inches b) between 58 inches and 62 inches.	10	CO3	BL3
4.	It is seen that 17.26% people lost their money in the stock market. In a certain year 640 people invested in a stock and 63 persons lost their money. Can the stock be considered as believable at 1% level of significance (LOS)? What about at 5% LOS?	10	CO4	BL3
5.	A bakery claims their new muffins have an average diameter of 5 centimetres (cm) with a standard deviation of 0.5 cm.	10	CO4	BL4

General instruction(s): Answer ALL Questions

Q. No.	Question	Max Marks	CO	BL																																				
1.	The sale of a product in lakhs of rupees(Y) is expected to be influenced by two variables namely the advertising expenditure(X_1)(in thousands of rupees) and the number of sales persons(X_2) in a region. Sample data on 8 regions of a state has given the following results: <table><tr><th>Area</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th></tr><tr><td>Y</td><td>105</td><td>95</td><td>80</td><td>115</td><td>135</td><td>100</td><td>95</td><td>125</td></tr><tr><td>X_1</td><td>20</td><td>33</td><td>38</td><td>25</td><td>28</td><td>24</td><td>27</td><td>33</td></tr><tr><td>X_2</td><td>13</td><td>15</td><td>7</td><td>9</td><td>6</td><td>12</td><td>14</td><td>11</td></tr></table> Find the regression model.	Area	1	2	3	4	5	6	7	8	Y	105	95	80	115	135	100	95	125	X_1	20	33	38	25	28	24	27	33	X_2	13	15	7	9	6	12	14	11	10	CO3	BL5
Area	1	2	3	4	5	6	7	8																																
Y	105	95	80	115	135	100	95	125																																
X_1	20	33	38	25	28	24	27	33																																
X_2	13	15	7	9	6	12	14	11																																
2.	Let an irregular 6-faced die is thrown and the possibility that in 10 throws it will give 5 even numbers is twice the expectation that it will give 4 even numbers. Out of 10,000 sets of 10 throws, find the number of times you should expect to receive (i) no even numbers, (ii) at least 15 even numbers and (iii) at most 3 even numbers.	10	CO2	BL3																																				
3.	Assume that there were 200 students participated in an annual examination in a school. As a result, their respective average score and standard deviation are 32 and 13. (i) How many candidates can be expected to obtain marks between 35 and 75 assuming the normality of the distribution and (ii) determine the limit of the marks of the central 60 % of the candidates.	10	CO2	BL2																																				

4.	A company wants to improve the quality of products by reducing defects and monitoring the efficiency of assembly lines. In assembly line A, there were 22 defects reported out of 400 samples while in line B, 33 defects out of 800 samples were noted. Is there a difference in the procedures at $\alpha=2\%$ LOS?	10	CO4
5.	a) A sample of 200 students is taken from a large population. The mean height of the students in this sample is 135cm. Can it be reasonably regarded that this sample is from a population of mean 135 cm and S.D 8 cm? Also find the 95% fiducial limits for the mean. b) A genetic experiment involving peas yielded one sample of offspring consisting of 334 green peas and 143 yellow peas. Use a 0.01 significance level to test the claim that under the same circumstances, 18% of offspring peas will be yellow. Identify the null hypothesis, alternative hypothesis, test statistic, P-value, conclusion about the null hypothesis, and final conclusion that addresses the original claim. Use the P-value method.	(5+5)	CO4 E

Exam Duration: 90 Min.

General instruction(s): Answer ALL Questions

(Statistical Table is to be permitted)

(Statistical Table is to be performed)

Q.No	Question	Max Marks																		
1.	Obtain the equations of the regression lines from the following data. Also estimate the values of (i) Y, when X=38 and (ii) values of X, when Y=18.	10																		
	<table border="1" style="margin-left: 40px;"> <tr> <td>X</td> <td>22</td> <td>26</td> <td>29</td> <td>30</td> <td>31</td> <td>31</td> <td>34</td> <td>35</td> </tr> <tr> <td>Y</td> <td>20</td> <td>20</td> <td>21</td> <td>29</td> <td>27</td> <td>24</td> <td>27</td> <td>31</td> </tr> </table>	X	22	26	29	30	31	31	34	35	Y	20	20	21	29	27	24	27	31	
X	22	26	29	30	31	31	34	35												
Y	20	20	21	29	27	24	27	31												
2.	In a certain factory producing razor blades, there is a small chance 0.002 for any blade to be defective. The blades are supplied in packets of 10. Use Poisson distribution to calculate the approximate number of packets containing (i) no defective blade, (ii) at least one defective blade and (iii) at most 1 defective blade in a consignment of 10000 packets.	10																		
3.	The lifetime X (in hundreds of hours) of a certain type of vacuum tube has a Weibull distribution with parameters $\alpha=2$ and $\beta=3$. Compute the following: (i) $E(X)$ and $Var(X)$, (ii) $P(X \leq 6)$, (iii) $P(1.8 \leq X \leq 6)$ and (iv) $P(X \geq 3)$.	10																		
4.	A random sample of 100 recorded deaths in the United States during the past year showed an average life span of 71.8 years. Assuming a population standard deviation of 8.9 years, does this seem to indicate that the mean life span today is greater than 70 years at 0.05 level of significance.	10																		
5.	In a referendum submitted to the student body at a university, 850 men and 566 women voted. 530 of the men and 304 of the women votes yes. Does this indicate a significant difference of the opinion on the matter between men and women students?	10																		

**VIT**Vellore Institute of Technology
(Approved by AICTE, Government of India, New Delhi, India)**SCHOOL OF ADVANCED SCIENCES**

Winter Semester 2023-2024

Continuous Assessment Test -II

Programme Name & Branch: B.Tech

Slot: B1+TB1 (Common)

Course Name & code: Probability and Statistics & BMAT202L

Exam Duration: 90 Min.

Maximum Marks: 50

General instruction(s): Answer ALL Questions

(Use of statistical table is allowed)

Q.No.	Question	Max Marks	CO	BL																																	
1.	The following table gives the test scores made by ten sales man on an intelligent test and the values of their weekly sales: <table><tr><th>Salesman</th><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th><th>F</th><th>G</th><th>H</th><th>I</th><th>J</th></tr><tr><th>Test scores</th><td>40</td><td>70</td><td>50</td><td>60</td><td>80</td><td>50</td><td>90</td><td>40</td><td>60</td><td>60</td></tr><tr><th>Sales (in thousands)</th><td>2.5</td><td>6.0</td><td>4.5</td><td>5.0</td><td>4.5</td><td>2.0</td><td>5.5</td><td>3.0</td><td>4.5</td><td>3.0</td></tr></table> Obtain the regression equation of sales on test scores. Also estimate the most probable value of weekly sales for a salesman whose test score is 70.	Salesman	A	B	C	D	E	F	G	H	I	J	Test scores	40	70	50	60	80	50	90	40	60	60	Sales (in thousands)	2.5	6.0	4.5	5.0	4.5	2.0	5.5	3.0	4.5	3.0	10	CO3	BL2
Salesman	A	B	C	D	E	F	G	H	I	J																											
Test scores	40	70	50	60	80	50	90	40	60	60																											
Sales (in thousands)	2.5	6.0	4.5	5.0	4.5	2.0	5.5	3.0	4.5	3.0																											
2.	The probability of a man scoring a penalty in a Hockey match is $\frac{1}{2}$. How many times he should be given a chance so that the probability of scoring successfully at least once is greater than $\frac{1}{4}$.	10	CO2	BL3																																	
3.	If the number of kilometers that a car can run before its battery wears out is exponentially distributed with an average value of 10,000 km and if the owner desires to take a 5000 km trip, what is the probability that he will be able to complete the trip without having to replace the car battery? Assume that the car has been used for some time. What is the probability, when the distribution is not exponential?	10	CO2	BL3																																	
4.	A cool drinks manufacturing company claims that its brand A cool drinks outsells its brand B by 8%. It is found that 42 out of a sample of 200 people prefer brand A and 18 out of another sample of 100 people prefer brand B. Test at 10% level of significance, whether the	10	CO4	BL3																																	

No.	Question	Marks	CO	BL																																				
1.	The sale of a product in lakhs of rupees(Y) is expected to be influenced by two variables namely the advertising expenditure(X_1)(in thousands of rupees) and the number of sales persons(X_2) in a region. Sample data on 8 regions of a state has given the following results: <table><tr><th>Area</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th></tr><tr><td>Y</td><td>105</td><td>95</td><td>80</td><td>115</td><td>135</td><td>100</td><td>95</td><td>125</td></tr><tr><td>X_1</td><td>20</td><td>33</td><td>38</td><td>25</td><td>28</td><td>24</td><td>27</td><td>33</td></tr><tr><td>X_2</td><td>13</td><td>15</td><td>7</td><td>9</td><td>6</td><td>12</td><td>14</td><td>11</td></tr></table> Find the regression model.	Area	1	2	3	4	5	6	7	8	Y	105	95	80	115	135	100	95	125	X_1	20	33	38	25	28	24	27	33	X_2	13	15	7	9	6	12	14	11	10	CO3	BL5
Area	1	2	3	4	5	6	7	8																																
Y	105	95	80	115	135	100	95	125																																
X_1	20	33	38	25	28	24	27	33																																
X_2	13	15	7	9	6	12	14	11																																
2.	Let an irregular 6-faced die is thrown and the possibility that in 10 throws it will give 5 even numbers is twice the expectation that it will give 4 even numbers. Out of 10,000 sets of 10 throws, find the number of times you should expect to receive (i) no even numbers, (ii) at least 15 even numbers and (iii) at most 3 even numbers.	10	CO2	BL3																																				
3.	Assume that there were 200 students participated in an annual examination in a school. As a result, their respective average score and standard deviation are 32 and 13. (i) How many candidates can be expected to obtain marks between 35 and 75 assuming the normality of the distribution and (ii) determine the limit of the marks of the central 60 % of the candidates.	10	CO2	BL2																																				

4.	A company wants to improve the quality of products by reducing defects and monitoring the efficiency of assembly lines. In assembly line A, there were 22 defects reported out of 400 samples while in line B, 33 defects out of 800 samples were noted. Is there a difference in the procedures at $\alpha=2\%$ LOS?	10	CO4	BL3
5.	a) A sample of 200 students is taken from a large population. The mean height of the students in this sample is 135cm. Can it be reasonably regarded that this sample is from a population of mean 135 cm and S.D 8 cm? Also find the 95% fiducial limits for the mean. b) A genetic experiment involving peas yielded one sample of offspring consisting of 334 green peas and 143 yellow peas. Use a 0.01 significance level to test the claim that under the same circumstances, 18% of offspring peas will be yellow. Identify the null hypothesis, alternative hypothesis, test statistic, P-value, conclusion about the null hypothesis, and final conclusion that addresses the original claim. Use the P-value method.	(5+5)	CO4	BL4

No													
1.	The following tables gives the aptitude test scores(X) and productivity indices(Y) of 10 workers selected at random:										10	3	
	X	60	62	65	70	72	48	53	73	65	82		
	Y	68	60	62	80	85	40	52	62	60	81		
	a) Calculate productivity index of a worker whose test score is 92.												
	b) Calculate test score of a worker whose productivity index is 75.												
2.	A. (i) Bring out the fallacy if any, in the statement: "The mean of a binomial distribution is 28 and standard deviation is 6."											2	
	(ii) A machine produces on an average 20% defective bolts. A batch is accepted if a sample of 5 bolts taken from that batch contains no defective and rejected if the sample contains 3 or more defectives. In other cases, a second sample is taken. What is the probability that second sample is required.										2+ 3+ 5		
	B. If the probability that an individual suffers a bad reaction from an injection of a given serum is 0.001. Determine the probability that out of 2000 individual												
	(i) At most 2 will suffer a bad reaction.												
	(ii) At least 3 individual will suffer a bad reaction.												

Page 1 of 2



VIT
Vellore Institute of Technology
(Chartered as the University under section 3 of UGC Act, 1956)

**SCHOOL OF ADVANCED SCIENCES
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025**

REG.NO.:

SLOT: D1+ TD1

3.	In an examination it is laid down that a student passes if he scores 35 percent or more marks. He is placed in first, second or third division according as he scores 60% or more marks, between 45% and 60% marks, and marks between 35% and 45%, respectively. He gets distinction in case he secures 75% or more marks. It is noticed from the result that 15% of the students failed in the examination whereas 10% of them obtained distinction. Assuming normal distribution of marks, obtain (i) Mean and standard deviation of the distribution (ii) Calculate the percentage of the students got third division in the examination.	10	2	2
4.	The mean height of 50 male students who showed above average participation in college athletics were 68.2 inches with a standard deviation of 2.5 inches, while other 50 male students who showed no interest in such participation had a mean height of 67.5 inches with a standard deviation of 2.8 inches. Test the hypothesis that male students who participate in college athletics are taller than other male students, using 5% level of significance.	10	4	3
5.	1000 apples kept under one type of storage found to show rotting to the extent of 4%. 1500 apples kept under another kind of storage showed 3% rotting. Can it be reasonably concluded that the second type of storage is superior to the first. Test at 2% level of significance.	10	4	3

1.	Find the linear relation of Sales on Advertisement expenditure and Advertisement expenditure on Sales from the data given below and hence estimate: 1. The Sale for advertising expenditure of Rs. 90 lakhs 2. The advertisement expenditure for sales of Rs. 45 crores <table><tr><td>Sales (Rs.Crores)</td><td>13</td><td>16</td><td>22</td><td>20</td><td>24</td><td>30</td><td>35</td></tr><tr><td>Adv. Exp. (Rs.Lakhs)</td><td>51</td><td>64</td><td>65</td><td>71</td><td>76</td><td>80</td><td>74</td></tr></table>	Sales (Rs.Crores)	13	16	22	20	24	30	35	Adv. Exp. (Rs.Lakhs)	51	64	65	71	76	80	74	10	CO 3	BL3
Sales (Rs.Crores)	13	16	22	20	24	30	35													
Adv. Exp. (Rs.Lakhs)	51	64	65	71	76	80	74													
2.	With usual notation, find p for a binomial random variate X if $n = 6$ and if $9P(X = 4) = P(X = 2)$. Hence find $P(X = 1)$.	10	CO 2	BL3																
3.	The height X , of young Indian women is distributed normal with mean $\mu = 65.5$ and standard deviation. $\sigma = 2.5$ inches. Find (i) $P(X < 67)$ (ii) $P(64 < X < 67)$ and (iii) $P(X > 65)$.	10	CO 2	BL3																
4.	a) A group of cookery students is comparing two methods for preparing a dish: steaming and frying method. They want to know if patrons of their restaurant prefer their frying method over the steaming method. A sample of patrons are given the dish prepared using each method and asked to select their preference. A statistical	5	CO 4	BL4																

	analysis is performed to determine if more than 50% of participants prefer the new frying method: $H_0 : p = 0.50, H_1 : p > 0.50$ Discuss Type I and Type II error and their consequences for this scenario. b) A Telecom service provider claims that individual customers pay on an average Rs. 400. per month with standard deviation of Rs. 25. A random sample of 50 customers' bills during a given month is taken with a mean of Rs. 250 and standard deviation of Rs. 15. Test the hypothesis against the claim made by the service provider. Assume LOS 1%.	5		
5.	Random samples of 400 men and 600 women were asked whether they would like to have a flyover near their residence. 200 men and 325 women were in favour of the proposal. Test the hypothesis that proportions of men and women in favour of the proposal are same or not. Assume 5% LOS.	10	CO 4	BL4

No																									
1.	The following tables gives the aptitude test scores(X) and productivity indices(Y) of 10 workers selected at random: <table border="1"><tr><td>X</td><td>60</td><td>62</td><td>65</td><td>70</td><td>72</td><td>48</td><td>53</td><td>73</td><td>65</td><td>82</td></tr><tr><td>Y</td><td>68</td><td>60</td><td>62</td><td>80</td><td>85</td><td>40</td><td>52</td><td>62</td><td>60</td><td>81</td></tr></table> a) Calculate productivity index of a worker whose test score is 92. b) Calculate test score of a worker whose productivity index is 75.	X	60	62	65	70	72	48	53	73	65	82	Y	68	60	62	80	85	40	52	62	60	81	10	3 1
X	60	62	65	70	72	48	53	73	65	82															
Y	68	60	62	80	85	40	52	62	60	81															
2.	A. (i) Bring out the fallacy if any, in the statement: "The mean of a binomial distribution is 28 and standard deviation is 6." (ii) A machine produces on an average 20% defective bolts. A batch is accepted if a sample of 5 bolts taken from that batch contains no defective and rejected if the sample contains 3 or more defectives. In other cases, a second sample is taken. What is the probability that second sample is required. B. If the probability that an individual suffers a bad reaction from an injection of a given serum is 0.001. Determine the probability that out of 2000 individual (i) At most 2 will suffer a bad reaction. (ii) At least 3 individual will suffer a bad reaction.	2+ 3+ 5	2 2																						

Page 1 of 2



VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

SCHOOL OF ADVANCED SCIENCES
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

REG.NO.:

SLOT: D1+ TD1

3.	In an examination it is laid down that a student passes if he scores 35 percent or more marks. He is placed in first, second or third division according as he scores 60% or more marks, between 45% and 60% marks, and marks between 35% and 45%, respectively. He gets distinction in case he secures 75% or more marks. It is noticed from the result that 15% of the students failed in the examination whereas 10% of them obtained distinction. Assuming normal distribution of marks, obtain	10	2	2
	(i) Mean and standard deviation of the distribution (ii) Calculate the percentage of the students got third division in the examination.			
4.	The mean height of 50 male students who showed above average participation in college athletics were 68.2 inches with a standard deviation of 2.5 inches, while other 50 male students who showed no interest in such participation had a mean height of 67.5 inches with a standard deviation of 2.8 inches. Test the hypothesis that male students who participate in college athletics are taller than other male students, using 5% level of significance.	10	4	3
5.	1000 apples kept under one type of storage found to show rotting to the extent of 4%. 1500 apples kept under another kind of storage showed 3% rotting. Can it be reasonably concluded that the second type of storage is superior to the first. Test at 2% level of significance.	10	4	3

Continuous Assessment Test –II

Programme Name & Branch : B.Tech

Slot : D1+TD1 (Common)

Course Name & Code : Probability and Statistics & BMAT202L

Exam Duration: 90 Min.

Maximum Marks: 50

General instruction(s): Answer ALL Questions

(Statistical Table is to be permitted)

Q.No	Question	Max Marks	CO	BL																		
1.	Obtain the equations of the regression lines from the following data. Also estimate the values of (i) Y, when X=38 and (ii) values of X, when Y=18. <table><tr><td>X</td><td>22</td><td>26</td><td>29</td><td>30</td><td>31</td><td>31</td><td>34</td><td>35</td></tr><tr><td>Y</td><td>20</td><td>20</td><td>21</td><td>29</td><td>27</td><td>24</td><td>27</td><td>31</td></tr></table>	X	22	26	29	30	31	31	34	35	Y	20	20	21	29	27	24	27	31	10	CO3	BL3
X	22	26	29	30	31	31	34	35														
Y	20	20	21	29	27	24	27	31														
2.	In a certain factory producing razor blades, there is a small chance 0.002 for any blade to be defective. The blades are supplied in packets of 10. Use Poisson distribution to calculate the approximate number of packets containing (i) no defective blade, (ii) at least one defective blade and (iii) at most 1 defective blade in a consignment of 10000 packets.	10	CO2	BL5																		
3.	The lifetime X (in hundreds of hours) of a certain type of vacuum tube has a Weibull distribution with parameters $\alpha=2$ and $\beta=3$. Compute the following: (i) E(X) and Var(X), (ii) P(X≤6), (iii) P(1.8≤X≤6) and (iv)P(X≥3).	10	CO2	BL2																		
4.	A random sample of 100 recorded deaths in the United States during the past year showed an average life span of 71.8 years. Assuming a population standard deviation of 8.9 years, does this seem to indicate that the mean life span today is greater than 70 years at 0.05 level of significance.	10	CO4	BL5																		

Page 1 of 2

5.	In a referendum submitted to the student body at a university, 850 men and 566 women voted. 530 of the men and 304 of the women votes yes. Does this indicate a significant difference of the opinion on the matter between men and women students?	10	CO4	BL5
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1.	A departmental store gives an in-service training to its salesmen, which is followed by a test. It is considering whether it should terminate the services of any salesman who does not do well in the test. The following data give test scores and sales made by 9 salesmen during a certain period.																							
	<table><tr><td>Test Scores</td><td>18</td><td>23</td><td>28</td><td>25</td><td>30</td><td>26</td><td>19</td><td>24</td><td>23</td></tr><tr><td>Sales(Rs. '000)</td><td>51</td><td>56</td><td>68</td><td>57</td><td>70</td><td>65</td><td>53</td><td>61</td><td>59</td></tr></table>	Test Scores	18	23	28	25	30	26	19	24	23	Sales(Rs. '000)	51	56	68	57	70	65	53	61	59	10		
Test Scores	18	23	28	25	30	26	19	24	23															
Sales(Rs. '000)	51	56	68	57	70	65	53	61	59															
	(i) Obtain the two regression equations																							
	(ii) If the firm wants a minimum sales value of Rs. 50,000, what is the test score that will justify the continuous of the services.																							
	(iii) Estimate the most probable sales volume of a salesman making a score of 32.																							
2.	An irregular 6 faced dice is thrown thirty times and the number of sixes seen is eight. If the dice is thrown a further twelve times find: (a) the probability that a six will occur exactly twice; (b) the expected number of sixes; (c) the variance of the number of sixes.	5	2	2																				
		5																						
3.	The Vellore Institute of Technology installed 10,000 electric led bulbs in the Vellore campus. If these bulbs have an average life of 1,000 burning hours with a standard deviation of 200 hours, how many bulbs might be expected to fail (i) in the first 800 burning hours? (ii) between 800 and 1200 burning hours? After how many burning hours would you expect (iii) 10% of the bulbs to fail? (iv) 10% of the bulbs to be still burning? Assume that the life of the bulbs is normally distributed.	10	2	3																				

Page 1 of 2



VIT

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

SCHOOL OF ADVANCED SCIENCES
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025

REG.NO.:

SLOT: D2+TD2

4.	A survey claims that 9 out of 10 doctors recommend aspirin for their patients with headaches. To test this claim, a random sample of 100 doctors is obtained. Of these hundred doctors, 82 indicate that they recommend aspirin. Is this claim accurate?	10	4	3
5.	A sample of 100 bulbs of brand A gave a mean lifetime of 1200 hours with a S.D. of 70 hours, while another sample of 120 bulbs of brand B gave a mean lifetime of 1150 hours with S.D. of 85 hours. Can we conclude that brand A bulbs are superior to brand B bulbs?	10	4	3



VIT
Vellore Institute of Technology
Established by the Government of Tamil Nadu in 1984

**SCHOOL OF ADVANCED SCIENCES
CONTINUOUS ASSESSMENT TEST - II
WINTER SEMESTER 2024-2025**

REG.NO.:

SLOT: G1+TG1

	contaminated? (c) How many boxes must be selected so that the probability of having at last one contaminated box is larger than 0.75?			
3.	a) The life of a power transmission tower is exponentially distributed, with mean life 25 years. If three towers, operated independently, are being erected at the same time, what is the probability that at least 2 will still stand after 35 years?	5	2	3
	b) In a recent study on world happiness, participants were asked to evaluate their current lives on a scale from 0 to 10, where 0 represents the worst possible life and 10 represents the best possible life. The responses were normally distributed, with a mean of 5.4 and a standard deviation of 2.2. Find the probability that a randomly selected study participant's response was: (a) Less than 4 (b) Between 4 and 6?	5		
4.	Two types of medication for hives are tested to determine if there is a difference in the proportions of adult patient reactions. In a sample of 200 adults given medication A, 20 still had hives 30 minutes after taking the medication. In a sample of 200 adults given medication B, 12 still had hives 30 minutes after taking the medication. At the 1% significance level, is there a difference in the proportion of adults who still have hives 30 minutes after taking medications?	10	4	3
5.	In a certain factory there are two independent processes of manufacturing the same item. The average weight in a sample of 250 items produced from one process is found to be 120 ozs with a S.D of 12 ozs. While the corresponding figures in a sample of 400 items from the other process are 124 and 14. Find the standard Error of difference between the two sample means. Is this difference significant? Also find the 99% confidence limits for the difference in the average weights of items produced by the two processes respectively.	10	4	3

Answer all questions

Q. No	Description	Marks	CO	BT Level
1.	The regression lines of y on x and of x on y are respectively $y = ax + b$ and $x = cy + d$, where $c \neq 0$ and $ac \neq 1$. (i) Find the mean of x and y. (ii) Determine the value of correlation coefficient between x and y. (iii) Show that the ratio of the standard deviations of y and x is $\sqrt{\frac{a}{c}}$.	10	3	2
2.	(a) The correlation between anxiety (A) and academic achievement (B) is -0.369, intelligence (C) and academic achievement (B) is 0.918, and anxiety (A) and intelligence (C) is -0.245. Find the multiple correlation between academic achievement and the linear combination of intelligence and anxiety. [5 marks] (b) The average time between breakdowns of a generator is 15 days. (i) If the generator has just broken down, what is the probability that it will break down again within the next 21 days? (ii) What is the probability that the generator will run for at least 30 days without a breakdown? [5 marks]	10	3	2
3.	The average percentage of candidates in an examination is 42 with standard deviation of 10. If the minimum pass mark is 50% and 1000 candidates appear for the examination, (a) How many candidates can be expected to get pass mark? (b) If it is required that double the number of candidates should pass, what should be the minimum pass mark? (c) Find how many candidates are expected to score 70% or more?	10	2	3

4.	(a) After correcting 50 pages of the proof of a book, the proof reader finds that there are, on the average, 2 errors per 5 pages. How many pages would one expect to find with 0, 1, 2, 3 and 4 errors, in 1000 pages of the first print of the book? [5 marks] (b) If X and Y are two independent binomial variates with parameters $(n_1 = 3, p = 0.4)$ and $(n_2 = 4, p = 0.4)$ respectively, then find (i) $P(X + Y \leq 2)$ (ii) $P(X = 3 X + Y = 4)$ [5 marks]	10	2	4
5.	(a) A sample of 900 items has mean 3.4 and standard deviation 2.61. Can the sample be regarded as drawn from a population with mean 3.25 at 5% level of significance? [5 marks] (b) It is thought that over 60% of the business offices in the United States have a mainframe computer as part of their equipment. (i) Set up the appropriate null and alternative hypotheses for supporting this claim. (ii) Find the critical point for an $\alpha = 0.05$ level test. (iii) When data are gathered, it is found that 223 of 375 offices studied have mainframe computers. Can the null hypothesis be rejected at $\alpha = 0.05$ level? [5 marks]	10	4	3

The grades of a class of 8 students on a CAT report (X) and on the final examination (Y) are as follows

X	77	50	71	72	81	67	79	90
Y	82	66	78	54	47	76	75	65

- (a) Estimate the linear regression lines.
 - (b) Estimate the final examination grade of a student who received a grade of 88 on the CAT report.
2. A manufacturer knows that on average 20% of the electric bike produced require repairs with 1 year after they are sold. When 20 bikes are randomly selected, find the appropriate number x and y such that
- (a) The probability that at least x of them will require repairs is less than 0.5.
 - (b) The probability that at least y of them will not require repairs is greater than 0.8.
3. In a research study, it was determined that the survival time in week of an animal subject to a certain exposure of gamma radiation has a gamma distribution with $\alpha = 5$ and $\beta = 10$.
- (a) What is the probability that an animal survives more than 30 weeks?
 - (b) What is the probability that an animal survives less than 15 weeks?

(PT-8)



VIT

Vellore Institute of Technology

Vellore - 632014, Tamil Nadu, India

DEPARTMENT OF MATHEMATICS

SCHOOL OF ADVANCED SCIENCES

WINTER SEMESTER 2022-2023

4. Two groups A and B , each consist of 100 people who have a viral fever. A vaccine is given to Group A but not to Group B , the two groups are treated identically. It is found that in Groups A and B , 85 and 75 people, respectively, recover from the fever. Test the hypothesis that the vaccine helps to cure the fever using a level of significance of 1%. And also find the P value of the test.
5. A random sample of 64 bags of rice weighed, on average, 5.23 kg with a standard deviation of 0.24 kg. Test the hypothesis that $\mu = 5.5$ kg against the alternative hypothesis, $\mu < 5.5$ kg, at the 0.05 level of significance. Also find the 95% fiducial limits of the mean.